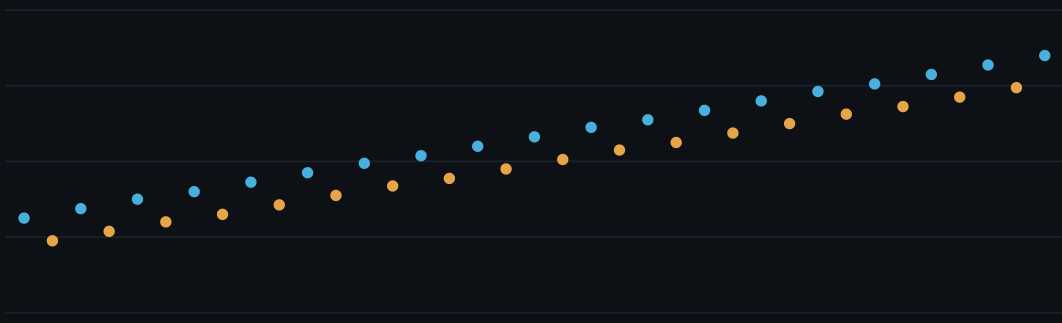
TicTac Grapher

User manual

Acoustic timegrapher for mechanical watches. Measure your watch's rate, amplitude, beat error and frequency using nothing but your phone's microphone.



Version 1.6 · October 2026 · Android 10 or later
Downloads: github.com/manursan2026/tictac-grapher

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> **Important note**
> TicTac Grapher is an indicative tool for enthusiasts. Its accuracy depends on the phone's microphone and internal clock and on ambient noise. It does not replace a professional timegrapher or a check by a watchmaker.

1 What is TicTac Grapher

A **timegrapher** (or "watch timing machine") is the instrument watchmakers use to check the condition of a mechanical watch without opening it. It listens to the ticking of the escapement and, from the exact moment of each sound, works out how the watch is behaving.

TicTac Grapher does the same with your phone's microphone. If you want to know exactly what produces that sound, see appendix A. The app measures:

Parameter	What it tells you
Rate (s/d)	Seconds the watch gains (+) or loses (–) per day.
Amplitude (°)	How many degrees the balance swings in each oscillation. It is the best indicator of the watch's "health": energy, lubrication and wear.
Beat error (ms)	Asymmetry between the "tick" and the "tock". It shows whether the balance is properly centred relative to the pallet fork.
Frequency (BPH)	Beats per hour of the movement (for example 21600 or 28800). It is detected automatically.

It also shows the **trace** (the classic timegrapher "paper tape"), the **waveform** of each beat, the **trend** of the readings over time and a log of measurements in **several positions**.

What you need

- A phone with Android 10 or later.
- A mechanical watch (hand-wound or automatic). Quartz watches will not work: their ticking is electronic.
- A quiet place and a soft surface: a cloth or a towel.
- Optional: a wired headset with a microphone, which often picks up the sound better.

Language

The app and its reports are available in **English and Spanish**. By default the app follows your phone's language (Spanish if the phone is set to Spanish, English otherwise). You can change it at any time in `Settings → Idioma / Language`.

Expected accuracy

- **Rate:** ±1–2 s/d, limited mainly by the accuracy of the phone's internal clock. You can compensate with the `Rate correction` setting.
- **Amplitude:** ±5–10°, provided the lift angle is set correctly.
- **Beat error:** ±0.1 ms.

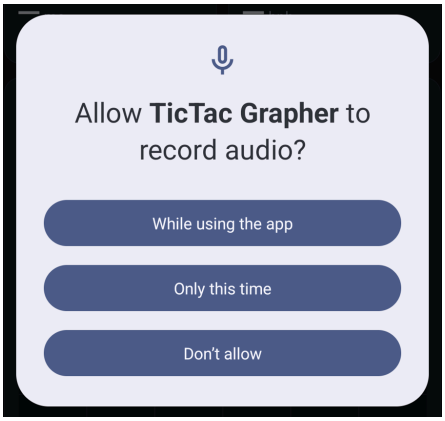
2 Installation

Installing the APK

1. On your phone, open the download page: `manursan2026.github.io/tictac-grapher`
2. Tap **Download APK** to download the file `TicTacGrapher.apk`.
3. Open the downloaded file. The first time, Android will ask you to allow "install unknown apps" for the browser or file manager: enable it and go back.
4. Tap **Install**. The app will appear as **TicTac Grapher**.

> **Updates**
> To update, download the APK of the new version and install it over the old one. Your settings and saved positions are kept.

Microphone permission



The first time you tap **▶ Measure**, the system asks for permission to use the microphone. Choose **"While using the app"**.

The audio is analysed **only on the phone**: the app does not send any sound to a server and does not need an Internet connection. A file is only saved if you use the **Record 20 s of audio** function.

If you denied the permission by mistake, enable it in *Phone settings → Apps → TicTac Grapher → Permissions → Microphone*.

3 The main screen



From top to bottom:

1. **Top bar.** `?` opens the quick measurement guide; `Demo` simulates a watch to try the app; `▶ Measure` / `■ Stop` starts or stops listening.
2. **Status line.** Coloured dot and message: **green** = measuring with a good signal, **yellow** = listening or insufficient signal. It shows the quality, the SNR and the spread (see chapter 5).
3. **Level bar** (to the right of the status line; on narrow screens it may be hidden). It moves with the sound reaching the microphone.
4. **Indicators.** Rate, amplitude, beat error and frequency. The colour shows whether the value is good, could be better or is poor. Below each value there is a detail line and the variation between readings (±).
5. **Trace.** One dot per beat: blue for tick, orange for tock. The buttons change the vertical scale.

Scrolling down you will find the **waveform**, the **trend**, the **positions** log and the **settings**.

Built-in quick guide

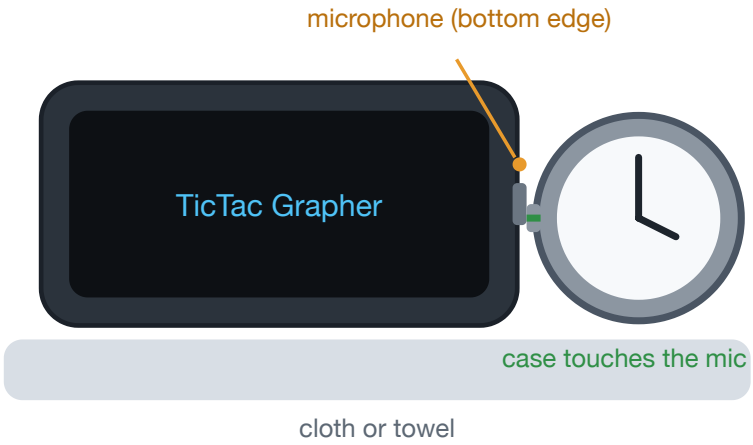
The first time you open the app, the guide **"How to measure your watch"** appears: a summary of this manual with illustrations. You can open it again at any time with the `?` button. On Android, the back button closes it.

Demo mode

`Demo` generates the sound of a simulated watch with known values, shown at the bottom of the screen. Use it to get familiar with the app and to check that it works without needing a watch.

4 How to take a measurement

1. **Prepare the watch.** Wind it fully: until it stops if hand-wound, or 30–40 turns of the crown if automatic. Let it run for a few minutes. To compare measurements from different days, always measure with the same state of wind (for example, right after winding).
2. **Find silence.** Turn off fans, TV and music. Put the phone on a cloth or a folded towel: a hard table transmits vibrations and knocks.
3. **Set the lift angle** of your calibre in **Settings** (chapter 8). If you don't know it, leave 52°.
4. **Put the watch in contact with the microphone** (see figure). This is the most important step.
5. **Tap** **▶ Measure** and don't touch anything. The status line shows the progress ("stabilizing NN %").
6. **Wait for it to finish.** With **automatic stop** (on by default), the measurement ends by itself when several consecutive readings agree, usually within 15–30 s. The phone vibrates and the **diagnosis** appears (chapter 9). If it does not stabilize within 90 s, it stops anyway and warns that the reading is unreliable.



Top view: phone lying flat and watch standing on its side against the bottom edge, with the metal case touching the microphone hole.

Tips for picking up the sound well

- The main microphone is usually a small hole on the **bottom edge** of the phone, next to the charging port.
- The watch's **metal case** must touch it: the side, near the crown, or the caseback. Not the strap or the crystal.
- Remove the phone cover if it blocks the microphone.
- If you cannot find the microphone, tap Measure and slide the watch along the bottom edge. Leave it where the level bar moves in time with the ticking.
- With a **wired headset with a microphone**, rest the mic on the watch's caseback: it usually gives the cleanest signal.

Stopping manually

You can tap ■ **Stop** at any time. If there are already at least three valid readings, the app closes the measurement with what it has and still shows the diagnosis. If you prefer to watch the watch continuously (for example, while regulating it), turn off automatic stop in Settings.

Avoid moving the watch while measuring

The app automatically discards stretches with handling noise (rubbing, knocks), but it needs at least 4 clean seconds to give a reading. Every time you move the watch, wait another 15–20 s.

5 Interpreting the results

The indicators

Indicator	Green	Yellow	Red
Rate	up to ±6 s/d	±6 to ±20 s/d	more than ±20 s/d
Amplitude	250° or more	200–250°	less than 200°
Beat error	up to 0.6 ms	0.6–1.2 ms	more than 1.2 ms

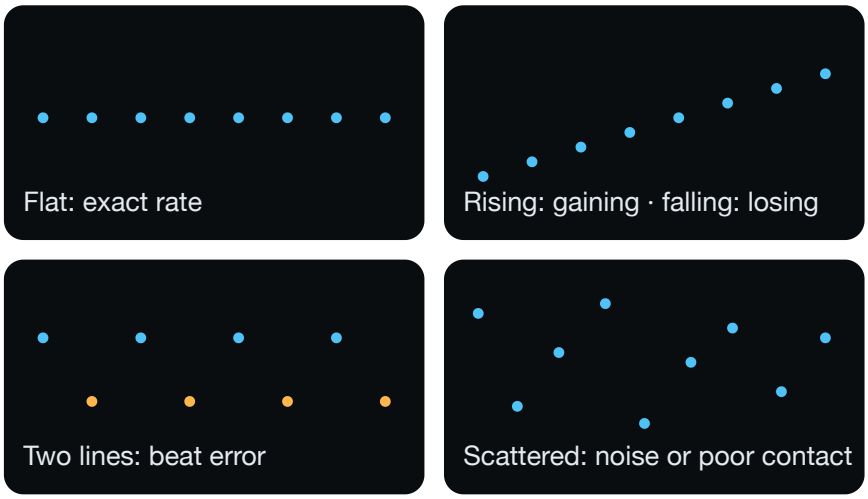
Each indicator shows the **median of the latest readings** (5 by default; one is taken every second). The small "±" value below shows how much they vary: the smaller, the more reliable the measurement. For the amplitude you will also see the **tick** and **tock** values separately.

The status line

Item	Meaning	Desirable value
Quality	Percentage of beats recognised and used in the calculation.	> 70 %
SNR	Signal-to-noise ratio: how much the ticking stands out from the background noise.	> 10 dB
Spread	Irregularity of the timing of each beat relative to the average rhythm.	< 0.3 ms

The trace (paper tape)

Each dot is a beat: blue for tick, orange for tock. The horizontal axis is time (the last 30 s) and the vertical axis is how early or late the beat is relative to the ideal rhythm.



Typical trace patterns.

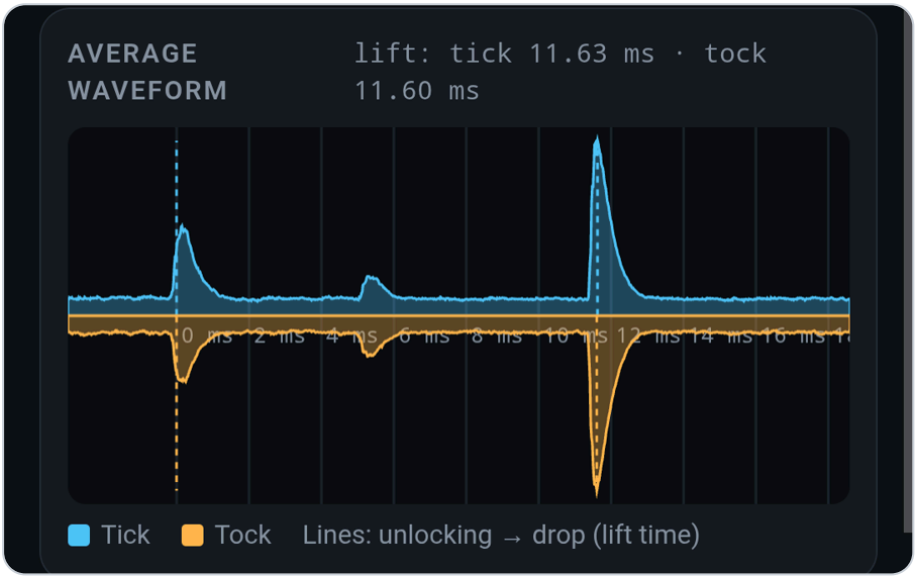
- **Scale.** `±2 ms`, `±5 ms` and `±20 ms` set the vertical range; `Period` shows a whole beat. If the line goes off the top or bottom, it reappears on the opposite side, as on a paper timegrapher.
- **A wavy line with a regular cycle** (for example every 60 s or every 7.5 s) is usually caused by an off-centre or damaged wheel in the gear train.

The average waveform

It shows the average sound of the tick (top, blue) and the tock (bottom, orange). In a healthy watch three sounds can be told apart in each beat (appendix A explains in detail where each one comes from):

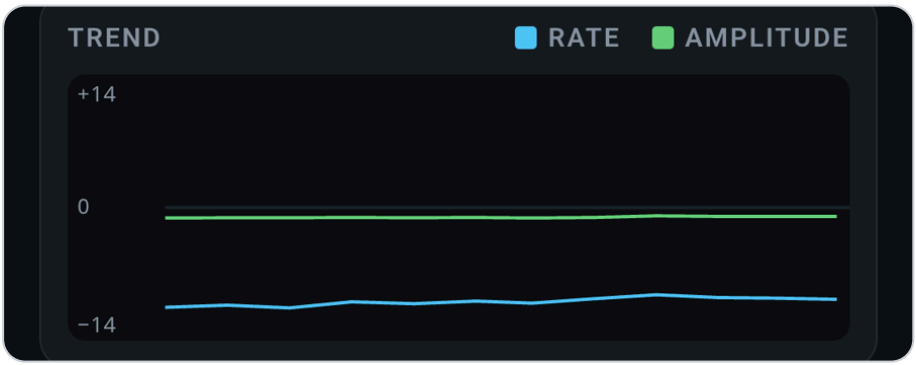
1. **Unlocking:** the balance's impulse jewel strikes the pallet fork. It is usually the weakest sound.
2. **Impulse:** an escape wheel tooth pushes the pallet.
3. **Drop:** the next tooth lands on the other pallet. It is usually the loudest.

The dashed lines mark the first and the last sound. The time between them is the **lift time** (top right), from which the amplitude is calculated.



Average waveform: the three escapement sounds and the lift time markers.

The trend



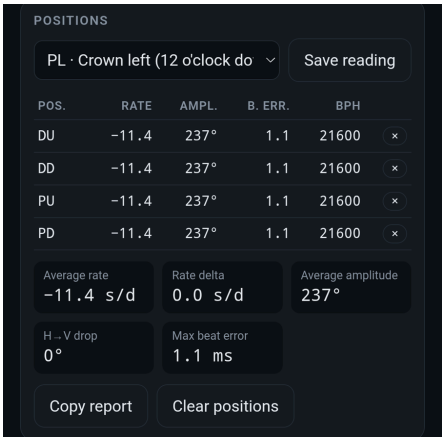
Rate (blue) and amplitude (green) over the session.

It lets you see whether the readings have settled and how rate and amplitude change when you turn the watch or as the mainspring unwinds.

6 Measuring in several positions

Gravity affects the balance, so a watch runs differently depending on its position. Watchmakers measure it in several positions to detect problems and fine-tune the regulation.

Code	Position
DU	Dial up (watch lying flat, crystal facing up)
DD	Dial down (crystal facing down)
PU	Crown up (on its side; 9 o'clock facing down)
PD	Crown down (on its side; 3 o'clock facing down)
PL	Crown left (on its side; 12 o'clock facing down)
PR	Crown right (on its side; 6 o'clock facing down)



1. Place the watch in the first position, with the case touching the microphone.
2. Wait **30 s** for the watch and the reading to settle.
3. Tap **Save reading**. The list moves on to the next position by itself.
4. Repeat for each position. With DU, DD and one or two vertical positions you already have a good diagnosis.

The summary calculates the **average rate**, the **rate delta** (maximum − minimum), the **average amplitude**, the **amplitude drop** between flat and vertical and the **maximum beat error**.

Copy report copies all the data as text so you can paste it into a message or a note. The × on each row deletes that position.

Summary	Good	Check
Rate delta between positions	< 10 s/d	> 25 s/d
Amplitude drop flat → vertical	20–40°	> 60°
Amplitude flat (fully wound)	270–310°	< 220°
Amplitude vertical (fully wound)	240–280°	< 200°

7 Settings



Settings are saved automatically on the phone and are kept between sessions and updates.

Setting	Default	What it is for and when to change it
Idioma / Language	Automatic	Language of the app and of the reports. «Automatic (device)» follows the phone's language: Spanish if the phone is in Spanish, English otherwise. Choose «Español» or «English» to fix it.
Lift angle	52°	Calibre figure needed to calculate the amplitude. See chapter 8.
Frequency (BPH)	Automatic	Set it manually only if the app detects half or double the real frequency (for example 14400 on a 28800 watch). This can happen with very large beat errors.
Averaging window	10 s	Seconds of audio used for each reading. A longer window (16–30 s) gives steadier readings but reacts more slowly to changes.
High-pass filter	3000 Hz	Removes low-pitched sounds (voices, hum, vibrations). Try 2000 Hz if the signal is weak or 4000–5000 Hz if there is a lot of low-pitched noise.
Rate correction	0 s/d	Compensates for the error of the phone's internal clock. If you compare with a professional timegrapher and the app always reads, say, 1.5 s/d too high, enter −1.5.
Average readings	5	Number of readings whose median is shown. Increase it (10–20) if the values fluctuate; decrease it (1–3) to see immediate reactions, for example while regulating the watch.
Automatic stop	Yes	Stops the measurement when the reading is stable and shows the diagnosis. Turn it off to measure continuously, for example while regulating the watch.
Freeze reading	No	Stops the values from updating so you can write them down or take a screenshot calmly.
Record 20 s of audio	—	Saves the microphone sound for analysis. See chapter 11.

8 The lift angle

The **lift angle** is the angle the balance turns through while it is in contact with the pallet fork. It is set by the design of the calibre. The app measures how long that contact lasts and, with this angle, calculates the amplitude:

$$\textbf{Amplitude} = \textbf{lift angle} \div (2 \cdot \sin(\pi \cdot t_{\textbf{lift}} \div T)), \text{ with } T = 7200 \div \text{BPH seconds.}$$

If the angle you set is not the one for your calibre, the **amplitude** will be off. An error of 1–2° changes the amplitude by about 4–8°. Rate and beat error do not depend on this setting.

Calibre	Frequency	Angle
ETA 2824, 2892, 7750 · Sellita SW200, SW300, SW500	28800	50°
Seagull ST2130 and ETA clones	28800	50–52°
Miyota 9015, 9039	28800	52°
Rolex 3135	28800	52°
Omega 8500 (co-axial)	25200	54°
Seiko NH35, NH36, 4R35, 4R36, 6R15, 7S26	21600	53°
Miyota 8215	21600	52°
Chinese «2813» and other Asian clones	21600	52° *
Unitas 6497/6498 and clones	18000	50°

* Approximate value: these makers rarely publish it.

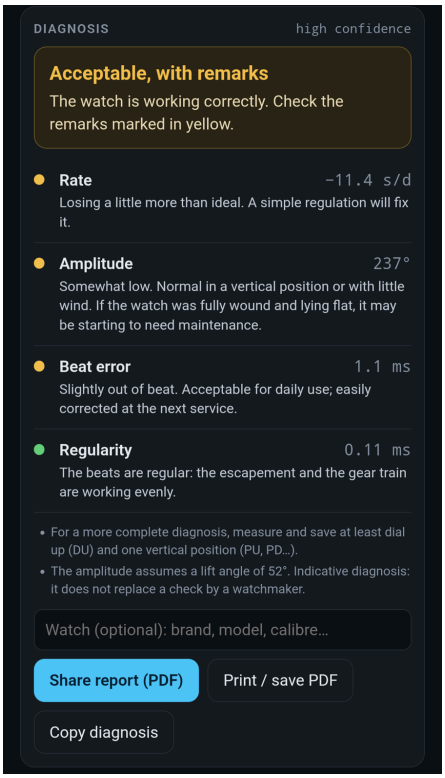
Don't know which calibre your watch has?

This is common with homages, replicas or watches without paperwork. You can narrow it down without opening it:

1. **Measure the frequency.** Tap Measure and note the BPH the app detects. Rule out calibres with a different frequency.
2. **Pull the crown all the way out** (time-setting position). If the **seconds hand stops**, the movement has hacking seconds: Seiko NH35/NH36, ETA 2824, Sellita SW200… If it **keeps running**, it points to a Miyota 8215 or a Chinese movement.
3. **Look at the rotor** if the caseback is transparent. It usually carries the maker's name: «TMI» or «Seiko Instruments» (NH35), «Miyota» or «Citizen», «ETA», «Sellita» or «SW», «Seagull»…
4. **Listen to the rotor.** Gently shake the watch next to your ear: the Miyota 8215 has a noisy rotor with a characteristic whirr.

If you still don't know, leave 52°: the conclusion (healthy watch or needs servicing) will not change because of an error of one or two degrees.

9 Diagnosis and PDF report



At the end of each measurement the **Diagnosis** card appears below the indicators:

- Overall **verdict**, with its explanation.
- **Remarks** for each parameter: rate, amplitude, beat error, regularity and, if you have saved positions, the difference between them and the amplitude drop. Each one has a coloured dot (**OK**, **attention**, **check**), its value and what it means.
- **Confidence** of the diagnosis (high, medium or low), based on how stable the reading was and on the signal quality.
- Notes: what is still to be measured, the lift angle assumed, etc.

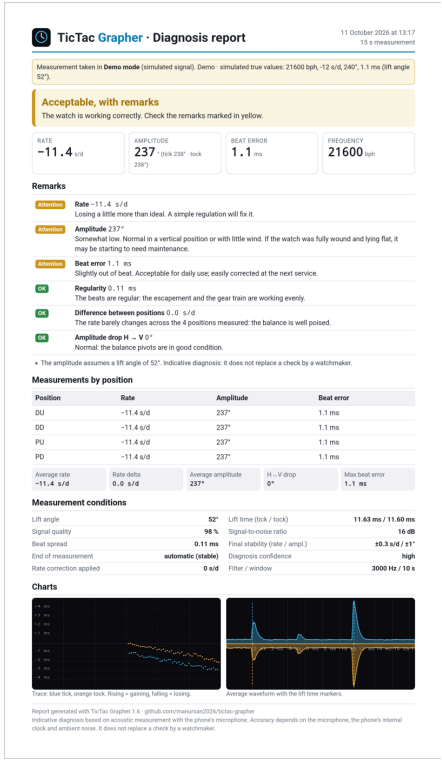
If you save positions afterwards, the diagnosis updates by itself.

Possible verdicts

Verdict	When it appears
Good condition	Every parameter is within the values of a healthy watch.
Acceptable, with remarks	It works well, but some value is in the attention zone.
Mechanically sound, needs adjusting	Amplitude and regularity are fine, but the rate or the beat error needs adjusting (regulation or beat).
Check: low amplitude	The amplitude is low (180–220°) and everything else is normal. Wind the watch fully and repeat: if it stays low, a service is advisable.
Probably magnetized	Extreme gain (more than +100 s/d) with every other parameter normal.
Needs servicing	Some indicator of the mechanical condition is poor: very low amplitude, irregular beats, large difference between positions or large amplitude drop.

> **Safeguards**
> With a weak signal (SNR below 8 dB), the amplitude is marked as "unreliable" and never produces the "Needs servicing" verdict. Regularity is only assessed with a clean signal (SNR of 12 dB or more), because with noise it would be measuring the noise rather than the watch.

Sharing or printing the PDF report


TicTac Grapher · Diagnosis report

11 October 2026 at 13:17
15 s measurement

Measurement taken in **Demo mode** (simulated signal). Demo · simulated true values: 21600 bph, -12 s/d, 240°, 1.1 ms (lift angle 52°).

Acceptable, with remarks
The watch is working correctly. Check the remarks marked in yellow.

RATE	AMPLITUDE	BEAT ERROR	FREQUENCY
−11.4 s/d	237 ° (tick 238° · tock 236°)	1.1 ms	21600 bph

Remarks

- Attention — **Rate** −11.4 s/d — Losing a little more than ideal. A simple regulation will fix it.
- Attention — **Amplitude** 237° — Somewhat low. Normal in a vertical position or with little wind. If the watch was fully wound and lying flat, it may be starting to need maintenance.
- Attention — **Beat error** 1.1 ms — Slightly out of beat. Acceptable for daily use; easily corrected at the next service.
- OK — **Regularity** 0.11 ms — The beats are regular: the escapement and the gear train are working evenly.
- OK — **Difference between positions** 0.0 s/d — The rate barely changes across the 4 positions measured: the balance is well poised.
- OK — **Amplitude drop H → V** 0° — Normal: the balance pivots are in good condition.

• The amplitude assumes a lift angle of 52°. Indicative diagnosis: it does not replace a check by a watchmaker.

Measurements by position

Position	Rate	Amplitude	Beat error
DU	−11.4 s/d	237°	1.1 ms
DD	−11.4 s/d	237°	1.1 ms
PU	−11.4 s/d	237°	1.1 ms
PD	−11.4 s/d	237°	1.1 ms

Average rate	Rate delta	Average amplitude	H → V drop	Max beat error
−11.4 s/d	0.0 s/d	237°	0°	1.1 ms

Measurement conditions

Lift angle	52°	Lift time (tick / tock)	11.63 ms / 11.60 ms
Signal quality	98 %	Signal-to-noise ratio	16 dB
Beat spread	0.11 ms	Final stability (rate / ampl.)	±0.3 s/d / ±1°
End of measurement	automatic (stable)	Diagnosis confidence	high
Rate correction applied	0 s/d	Filter / window	3000 Hz / 10 s

Charts

Trace: blue tick, orange tock. Rising = gaining, falling = losing.

Average waveform with the lift time markers.

Report generated with TicTac Grapher 1.6 · github.com/manunsan2026/tictac-grapher
Indicative diagnosis based on acoustic measurement with the phone's microphone. Accuracy depends on the microphone, the phone's internal clock and ambient noise. It does not replace a check by a watchmaker.

1. Optional: type the watch name (brand, model, calibre) in the field on the diagnosis card. It is remembered for next time and appears in the report and in the file name.
2. **To send it** via WhatsApp, email, Telegram, Drive…: tap **Share report (PDF)**. The app generates the PDF, saves a copy in `Downloads/TicTac` and opens the share menu. Choose the app and the contact: the PDF is attached, together with a short summary (verdict and values).
3. **To print or save it** with selectable text: tap **Print / save PDF**. In the printer drop-down choose **"Save as PDF"** or a printer.

The report includes: date and duration, watch name, verdict, the four indicators, all the remarks, the measurements by position with their summary, the measurement conditions (angle, quality, SNR, stability, confidence, settings used) and the trace and waveform charts. If the measurement was taken in Demo mode, the report says so. It is written in the language the app is set to.

The files are named `Report-TicTac-<watch>-YYYYMMDD-HHMM.pdf` (or `Informe-TicTac-…` in Spanish). **Copy diagnosis** copies the same content as text, to paste into a message.

What each symptom indicates

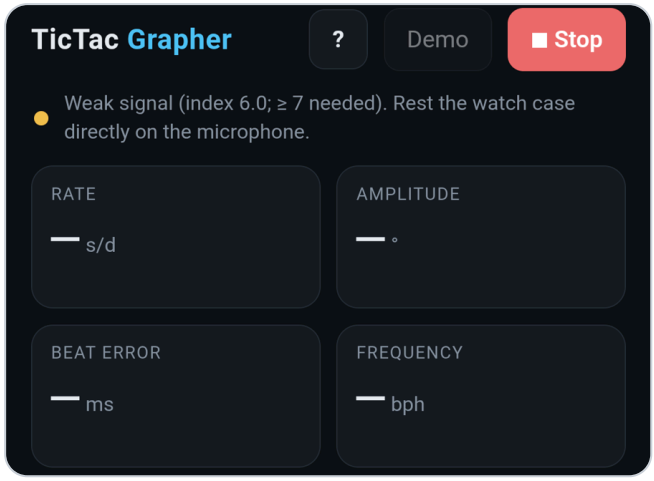
The table summarises the criteria the diagnosis is based on. They are general guidelines: the final diagnosis is a watchmaker's job.

Symptom	Likely causes	What to do
Low amplitude in all positions (< 200–220° fully wound)	Insufficient wind; dried-out or degraded oils; dirt; tired mainspring.	Wind fully and repeat. If it persists, it needs servicing (cleaning and lubrication).
Large amplitude drop between flat and vertical (> 60°)	Worn or damaged balance pivots; balance jewels or lubrication in poor condition.	Have a watchmaker check the balance.
Very high amplitude (> 330°)	Lift angle set incorrectly; mainspring too strong, with a risk of knocking.	Check the angle. If it is correct, consult a watchmaker.
High beat error (> 1.2 ms)	Balance out of beat relative to the pallet fork, for example after a knock or tampering.	Beat adjustment by a watchmaker. It is a simple operation.
Rate off but stable	Regulation.	Regulation by a watchmaker, or by yourself if you have experience (balance regulator).

Gains a huge amount (+100 s/d or more)	Watch **magnetized**: the hairspring coils stick together.	Demagnetize (many watch shops do it on the spot) and measure again.
Large rate difference between positions (> 25 s/d)	Balance out of poise; deformed hairspring; pivots in poor condition.	Service by a watchmaker.
Wavy trace with a regular cycle	Gear-train or escape wheel off-centre or with damaged teeth.	Service.
Very scattered trace with a good signal	Dirt or a damaged component in the escapement.	Service.

> **For a reliable diagnosis**
> Always measure fully wound and compare flat (DU, DD) with vertical (PU, PD…). Repeat the measurement after 24 hours without winding: the difference in amplitude shows the effective power reserve.

10 Troubleshooting



The status line shows at all times whether the app can hear the watch.

Message or symptom	What to do
"Listening… NN %"	Normal during the first few seconds or after moving the watch. Wait without touching anything.
"The ticking is not recognised"	The microphone cannot hear the watch. Check that the case touches the microphone hole, remove the cover and reduce the noise. Try the other edge of the phone (some models have the main microphone at the top) or a headset with a microphone.
"Weak signal (index X; ≥ 7 needed)"	Something can be heard, but not clearly enough. Improve the contact with the case, try the 2000 Hz filter or use a headset with a microphone.
"A rhythm can be heard, but the beats are not clean"	There is noise or knocking. Put the phone on something soft, don't touch the table and move away from noise sources.
The values change a lot	Increase **Average readings** to 10–20 or the **Window** to 16–30 s. Check that the SNR is above 10 dB.
Wrong BPH (half or double)	Set the frequency manually in Settings.
Absurd amplitude (very high or low, or tick and tock very different)	Check the lift angle. Try another filter (2000 or 4000 Hz). A watch whose escapement sounds are poorly defined can give unreliable amplitudes even if the rate is correct.
The rate does not match reality	The phone's internal clock may drift 1–2 s/d. Compensate with **Rate correction**. Also remember that the rate changes with position and state of wind.
"Could not open the microphone"	Grant the microphone permission (chapter 2) and close other apps that may be using it, such as calls or voice recorders.
The measurement does not stop by itself	The reading does not stabilize because of noise or poor contact: it will stop after 90 s with low confidence. Improve the placement and repeat. Also check that automatic stop is enabled in Settings.

WhatsApp does not appear when sharing	Swipe the share menu up to see all apps. The PDF is also saved in `Downloads/TicTac`: you can attach it from WhatsApp → Document.
"Save as PDF" does not appear	In the print dialog, tap the printer drop-down (at the top) and choose "Save as PDF". On some phones it is under "All printers".
The app is in the wrong language	Choose the language in **Settings → Idioma / Language**. «Automatic (device)» follows the phone's language.
The screen turns off	The app keeps the screen on while it is open. If it still turns off, check the phone's battery saver mode.

11 Diagnostic recording

If the app cannot measure your watch, you can record the sound exactly as the microphone picks it up so it can be analysed in detail and the app improved.

1. Place the watch on the microphone and tap **▶ Measure**.
2. In **Settings**, tap **Record 20 s of audio**. The button shows the countdown. Don't touch anything while it records.
3. When it finishes, the file is saved in `Downloads/TicTac` and the share menu opens (email, Drive, Quick Share…).

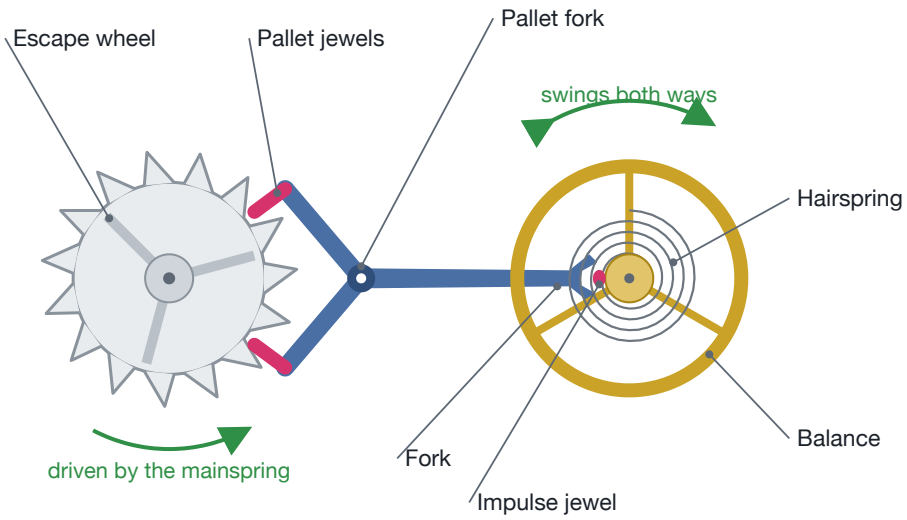
The file is an uncompressed WAV (`tictac-YYYYMMDD-HHMMSS.wav`, about 2 MB). Send it as a file: if you send it as a voice note, apps such as WhatsApp compress it and detail is lost.

> **Privacy**
> The recording captures everything the microphone hears for 20 seconds, including voices. Record it in silence.

A The escapement: where the ticking comes from

The "tick-tock" of a mechanical watch is the sound of the **escapement**, the mechanism that releases the energy of the mainspring in small, regular steps. Each step is a "tick" or a "tock", and that is what TicTac Grapher listens to.

The watch stores energy in the **mainspring**, the spring you tension when you wind it. If that energy were released all at once, the spring would unwind in seconds. The escapement releases it in identical steps, several times a second, and keeps the balance oscillating, which is what keeps time.



Simplified diagram of the Swiss lever escapement, used in almost every modern mechanical watch, seen from above.

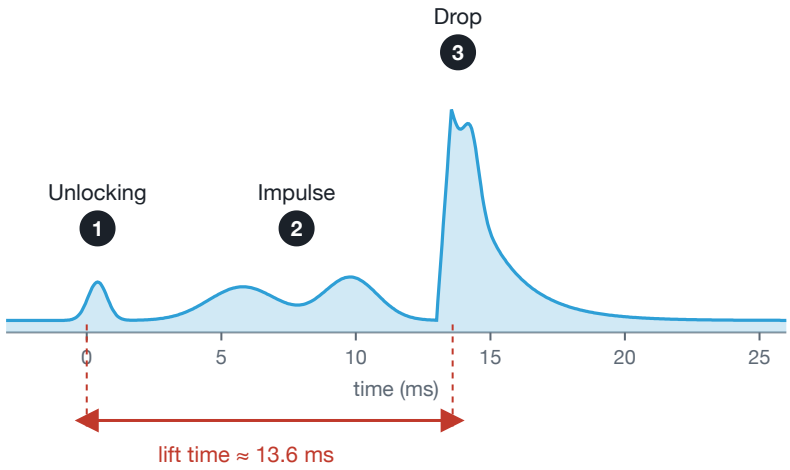
Part	Function
Escape wheel	Toothed wheel driven by the mainspring through the gear train. It always "wants" to turn.
Pallet fork	Anchor-shaped part that pivots on its arbor. Its two **pallet jewels** lock the escape wheel and release it one tooth at a time. Its other end, the **fork**, connects with the balance.
Balance	Wheel that swings back and forth at a very steady rate, like a pendulum. It carries the **impulse jewel** (also called the roller jewel), a small ruby that enters the fork.
Hairspring	Very fine spring that returns the balance to its rest position. Together with the balance, it sets the frequency of the watch.

What happens in each beat

While the balance swings freely, the pallet fork keeps the escape wheel locked. When the balance passes through the centre of its swing, the impulse jewel enters the fork and three things happen in quick succession. Each one makes a small click:

1. **Unlocking.** The jewel strikes the fork and turns the pallet fork. The pallet that was locking the escape wheel moves aside and releases it. It is usually the weakest sound.
2. **Impulse.** The escape wheel, driven by the spring, moves forward. A tooth slides along the inclined face of the pallet and pushes the pallet fork, which in turn pushes the balance through the jewel. This is how the balance recovers the energy it loses to friction.
3. **Drop.** The wheel keeps moving until another tooth hits the other pallet and is locked again. It is usually the loudest sound.

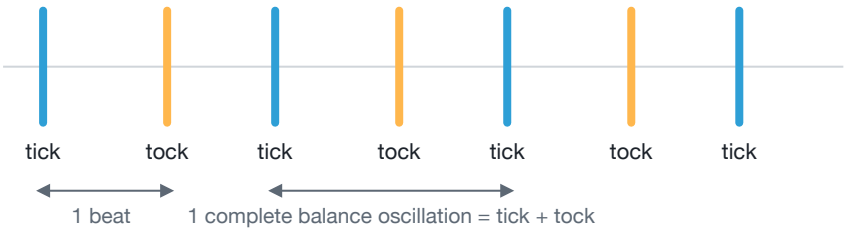
The balance then keeps swinging freely to the end of its travel, stops, the hairspring brings it back and the process repeats in the opposite direction.



The three sounds of a real beat, reconstructed from a recording of a 21600 bph watch. This is what the app's "Average waveform" chart shows.

Why it goes "tick" and then "tock"

The balance swings in both directions and the escapement acts in both: on the way out it produces one beat (the tick) and on the way back, another (the tock). One complete oscillation of the balance is therefore **two beats**.



Tick and tock alternate at equal intervals if the watch is in beat.

Frequency and beats

Frequency	Beats per second	Oscillations per second	Duration of a beat
18000 bph	5	2.5 Hz	200 ms
21600 bph	6	3 Hz	167 ms
28800 bph	8	4 Hz	125 ms
36000 bph	10	5 Hz	100 ms

What the app measures from the ticking

Measurement	How it is obtained from the sound
Rate	It compares the moment of each beat with the ideal rhythm for its frequency. If the beats arrive slightly earlier each time, the watch is gaining; if later, it is losing. A difference of just 1 microsecond per beat at 28800 bph equals 0.7 s/d.
Beat error	It compares the tick → tock and tock → tick intervals. If they differ, the impulse jewel is not centred in the fork when the balance is at rest, and the balance receives its impulses asymmetrically.
Amplitude	It measures the **lift time**: from sound 1 to sound 3, which is how long the balance takes to travel through its calibre's lift angle. The greater the amplitude, the faster the balance passes through the centre and the shorter that time. For example, 13.6 ms at 21600 bph with 52° is about 200°; at 270° it would be about 10 ms.
Regularity	It measures how much each beat deviates from the average rhythm. A clean escapement in good condition produces very regular beats; dirt or wear make them irregular.

What the sound gives away

- **Weak or blurred sounds** in the waveform: dried-out oils, dirt or worn pallets.
- **Tick and tock with very different shapes:** escapement asymmetry (beat error or wear of one pallet).
- **Low amplitude:** too little energy reaches the balance (weak mainspring, friction) or the balance loses it (worn pivots, degraded oils).
- **Excessive amplitude:** the balance swings so far that the jewel can strike the back of the fork (*knocking*), with a sudden gain in rate.

B Glossary

Amplitude
Maximum angle the balance swings to each side of its rest position. A healthy, fully wound watch is usually between 250° and 310°.

Balance
Oscillating wheel that, together with the hairspring, sets the rhythm of the watch.

Beat error
Difference in duration between tick and tock, in milliseconds. Ideally 0.

BPH (beats per hour)
28800 BPH = 8 beats per second = 4 Hz. Each complete oscillation of the balance produces two beats: tick and tock.

Calibre
Model of the watch's movement (the mechanism), for example ETA 2824 or Seiko NH35.

Escapement
Escape wheel and pallet fork, which deliver the mainspring's energy to the balance in regular impulses. It produces the sound the app analyses (see appendix A).

Hairspring
Very fine spring attached to the balance that makes it oscillate. If it becomes magnetized or deformed, the rate changes a lot.

Impulse jewel (roller jewel)
Small ruby fixed to the balance that enters the pallet fork in each beat.

Lift angle
Angle the balance travels through while in contact with the pallet fork. It is a fixed characteristic of each calibre.

Lift time
Time between unlocking and drop. The app uses it to calculate the amplitude.

Mainspring
Spring that stores energy when the watch is wound.

Pallet fork
Escapement part that, driven by the balance, alternately locks and releases the escape wheel.

Positions (DU, DD, PU, PD, PL, PR)
Standard orientations in which a watch is measured. D = dial, P = pendant (crown); U = up, D = down, L/R = left/right.

Rate
Daily gain (+) or loss (–) of the watch, in seconds per day (s/d).

SNR
Signal-to-noise ratio, in decibels. The higher it is, the cleaner the pickup.

Timegrapher
Instrument that acoustically measures how a mechanical watch is running.

Unlocking, impulse and drop
The three sounds that make up each beat (see appendix A).

C How the measurement works

For the curious, this is the process the app follows with the microphone's sound:

1. **Filtering.** The audio (44.1 or 48 kHz) goes through a band-pass filter (3–14 kHz by default) that keeps the metallic "clicks" of the escapement and removes voices, hum and vibrations.
2. **Noise gate.** 50 ms stretches whose background noise rises sharply, such as when the watch is handled, are discarded.
3. **Frequency.** The autocorrelation of the envelope reveals the period of the ticking. The standard frequencies (12000 to 43200 BPH) are tried and the best match is chosen.
4. **Beats.** Each beat is located on a time grid. Its position is then refined to a fraction of a sample by comparing it with a template of the typical sound: the median of all the beats, which is immune to isolated knocks.
5. **Rate and beat error.** A least-squares fit of the times of all the beats gives the real period (and hence the rate) and the offset between tick and tock (the beat error). Anomalous beats are discarded.
6. **Amplitude.** In each beat's template the unlocking (the first sound) and the drop (the dominant sound) are located. The time between them, together with the lift angle and the frequency, gives the amplitude.
7. **Display.** This is repeated every second with the last 10 s of audio and the median of the latest readings is shown.
8. **Automatic stop.** The measurement ends when six consecutive readings fall within ±1.5 s/d, ±5° and ±0.15 ms (after at least 10 s of valid signal). The final result is their median, which is turned into the diagnosis using rules based on the reference values in chapter 9.

All the calculation happens on the phone itself; the sound never leaves the device.